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JEE Main Physics Practice Sheet — DPP-041

Topic: Conservation of Mechanical Energy — Derivations, Curves & Key Applications

Time Allowed: 60 Minutes | Max Marks: 80 | Marking Scheme: +4, -1

Q1. A particle of mass m moves in an isolated system under a conservative force field $\vec{F}$. If W_c is the work done by the conservative force, the fundamental derivation of mechanical energy conservation combines:

- (A) $W_c = \Delta K$ and $W_c = -\Delta U \implies \Delta(K + U) = 0$
 (B) $W_c = -\Delta K$ and $W_c = \Delta U \implies \Delta(K - U) = 0$
 (C) $W_c = \Delta U$ and $W_{\text{ext}} = 0 \implies K = U$
 (D) $W_{\text{net}} = \Delta K$ and $W_{\text{friction}} = 0 \implies K + U = 0$

Q2. A small body of mass m slides down from the top of a smooth fixed hemispherical dome of radius R . The height h above the horizontal base at which the body loses contact with the dome is:

- (A) $\frac{1}{2}R$
 (B) $\frac{2}{3}R$
 (C) $\frac{3}{4}R$
 (D) $\frac{3}{4}R$

Q3. A bob of mass m attached to a light inextensible string of length L is projected horizontally at the lowest point with speed $u = \sqrt{3gL}$. The angle θ with the upward vertical at which the string becomes slack is:

- (A) 30°
 (B) 45°
 (C) 60°
 (D) 90°

Q4. A block of mass $m = 1$ kg is dropped from a height $h = 45$ cm onto a vertical spring of stiffness $k = 1000$ N/m. The maximum compression x of the spring is ($g = 10$ m/s²):

- (A) 0.05 m
 (B) 0.15 m
 (C) 0.20 m
 (D) 0.10 m

Q5. A particle of mass $m = 2$ kg moves along the x -axis in a conservative field with potential energy $U(x) = (2x^2 - 8x + 10)$ J. If the total mechanical energy is $E = 18$ J, the range of motion of the particle is between:

- (A) $x = 0$ m and $x = 4$ m
 (B) $x = -1$ m and $x = 5$ m
 (C) $x = -2$ m and $x = 6$ m
 (D) $x = 1$ m and $x = 3$ m

Q6. A light elastic cord of unstretched length L and spring constant k has one end fixed at O . A block of mass m attached to the other end is dropped from O . The maximum downward displacement of the block before it momentarily comes to rest is:

- (A) $L + \frac{mg}{k} + \sqrt{\frac{2mgL}{k}}$
 (B) $L + \frac{mg}{k} + \sqrt{\frac{m^2g^2}{k^2} + \frac{2mgL}{k}}$
 (C) $L + \frac{2mg}{k}$

(D) $\frac{mg}{k} + \sqrt{\frac{2mgL}{k}}$

Q7. A block of mass m released from rest slides down a smooth curved track and loops around a vertical circular loop of radius R . The minimum release height h above the bottom of the loop to complete the loop safely is:

- (A) $2R$
 (B) $2.5R$
 (C) $3R$
 (D) $1.5R$

Q8. A chain of mass M and length L is held vertically with its lower end just touching the floor and released from rest. The normal force exerted by the floor on the fallen chain when a length y has accumulated is:

- (A) $\frac{Mgy}{L}$
 (B) $\frac{2Mgy}{L}$
 (C) $\frac{3Mgy}{L}$
 (D) $\frac{4Mgy}{L}$

Q9. A particle is projected from the ground with speed u at an angle θ to the horizontal. By conservation of mechanical energy, the speed of the particle when it reaches half of its maximum height is:

- (A) $u\sqrt{1 - \frac{\sin^2 \theta}{2}}$
 (B) $u \cos \theta$
 (C) $\frac{u}{\sqrt{2}}$
 (D) $u\sqrt{\frac{1 + \cos^2 \theta}{2}}$

Q10. Two blocks of masses m_1 and m_2 are connected by a spring of stiffness k resting on a smooth floor. If m_1 is given a sudden velocity v_0 directed towards m_2 , the maximum compression of the spring is:

- (A) $v_0 \sqrt{\frac{m_1 m_2}{(m_1 + m_2)k}}$
 (B) $v_0 \sqrt{\frac{m_1}{k}}$
 (C) $v_0 \sqrt{\frac{m_1 + m_2}{k}}$
 (D) $v_0 \sqrt{\frac{m_1^2}{m_2 k}}$

Q11. A simple pendulum of length L and bob mass m is released from rest at an angle $\theta_0 = 60^\circ$ to the vertical. The tension in the string as the bob passes through the lowest position is ($g = 10$ m/s²):

- (A) mg
 (B) $2mg$
 (C) $3mg$
 (D) $1.5mg$

Q12. A particle moves in a 1D potential well given by $U(x) = \frac{a}{x^2} - \frac{b}{x}$ ($a, b > 0$). The escape speed of the particle from the bottom of the potential well (mass m) is:

- (A) $\frac{b}{\sqrt{2ma}}$
- (B) $\frac{b}{2\sqrt{ma}}$
- (C) $\frac{b}{\sqrt{ma}}$
- (D) $\frac{2b}{\sqrt{ma}}$

Q13. Under which of the following conditions is the total mechanical energy of a system of particles strictly conserved?

- (A) When the net external force on the system is zero
- (B) When only conservative internal/external forces do work ($W_{nc} = 0$)
- (C) When the linear momentum of the system is conserved
- (D) When the internal forces between particles cancel out

Q14. A uniform solid sphere of mass M and radius R is released from rest from height h on a smooth incline. Its speed at the bottom is v_1 . If the incline is rough enough to allow pure rolling without slipping, its speed at the bottom is v_2 . The ratio v_2/v_1 is:

- (A) $\sqrt{\frac{5}{7}}$
- (B) $\sqrt{\frac{7}{5}}$
- (C) $\sqrt{\frac{2}{5}}$
- (D) $\sqrt{\frac{5}{2}}$

Q15. A block of mass m is pushed against a horizontal spring of stiffness k resting on a smooth surface, compressing it by x_0 . A second block of mass $2m$ is placed touching the first. When released, the speed of the separation of the blocks when the spring regains natural length (spring attached to mass m only) is:

- (A) $x_0 \sqrt{\frac{k}{3m}}$
- (B) $x_0 \sqrt{\frac{k}{2m}}$
- (C) $x_0 \sqrt{\frac{2k}{3m}}$
- (D) $x_0 \sqrt{\frac{k}{m}}$

Q16. A small ring of mass m slides without friction on a vertical circular hoop of radius R . The ring is released

from rest at the top of the hoop. The normal reaction exerted by the hoop on the ring when it reaches the bottom is:

- (A) $3mg$
- (B) $4mg$
- (C) $5mg$
- (D) $6mg$

Q17. A body of mass m is moving on a frictionless horizontal table. It is attached to a light cord passing through a smooth hole in the center of the table. If the tension in the cord is increased by pulling it down slowly so that the radius of the circle decreases from r_1 to r_2 , the work done by the pulling force is:

- (A) $\frac{1}{2}mv_1^2 \left(\frac{r_1^2}{r_2^2} - 1 \right)$
- (B) $\frac{1}{2}mv_1^2 \left(\frac{r_1}{r_2} - 1 \right)$
- (C) $\frac{1}{2}mv_1^2 \left(1 - \frac{r_2^2}{r_1^2} \right)$
- (D) Zero

Q18. A small bead of mass m is threaded on a smooth parabolic wire $y = kx^2$ placed in a vertical plane (y -axis upward). If the bead is released from rest at height $y = h$, the magnitude of the normal reaction on the bead at the lowest point ($x = 0$) is:

- (A) $mg(1 + 4kh)$
- (B) $mg(1 + 2kh)$
- (C) $2mgkh$
- (D) $mg(1 - 2kh)$

Q19. A block of mass m is attached to two identical springs of constant k connected in parallel. If the block is displaced by x_0 and released, its kinetic energy at displacement $x = x_0/2$ is:

- (A) $\frac{3}{4}kx_0^2$
- (B) $\frac{3}{8}kx_0^2$
- (C) $\frac{1}{2}kx_0^2$
- (D) $\frac{1}{4}kx_0^2$

Q20. A pendulum string of length L is held horizontal and released. A nail is located at a distance d vertically below the suspension point. The minimum value of d so that the bob can describe a complete circle around the nail is:

- (A) $\frac{2}{5}L$
- (B) $\frac{3}{5}L$
- (C) $\frac{4}{5}L$
- (D) $\frac{1}{2}L$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-041)

Conservation of Mechanical Energy — Derivations, Curves & Key Applications

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	B	11	B	16	C
2	B	7	B	12	A	17	A
3	C	8	C	13	B	18	A
4	D	9	A	14	A	19	A
5	C	10	A	15	A	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)

By Work-Energy Theorem:

$$W_c + W_{nc} = \Delta K$$

For an isolated system with only conservative forces ($W_{nc} = 0$):

$$W_c = \Delta K$$

By definition of potential energy:

$$W_c = -\Delta U$$

Equating both expressions:

$$\Delta K = -\Delta U \implies \Delta(K + U) = 0 \implies K + U = \text{constant}$$

Sol 2. Option (B)

Let θ be angle with the vertical. Height dropped = $R(1 - \cos \theta)$.

By mechanical energy conservation:

$$\frac{1}{2}mv^2 = mgR(1 - \cos \theta) \implies v^2 = 2gR(1 - \cos \theta)$$

Normal force on dome:

$$mg \cos \theta - N = \frac{mv^2}{R} \implies N = mg(3 \cos \theta - 2)$$

Contact is lost when $N = 0 \implies \cos \theta = \frac{2}{3}$.

Height above base:

$$h = R \cos \theta = \frac{2}{3}R$$

Sol 3. Option (C)

Let string become slack at angle θ to the upward vertical.

Height from bottom: $H = L + L \cos \theta = L(1 + \cos \theta)$.

By conservation of mechanical energy:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgL(1 + \cos \theta)$$

$$3gL = v^2 + 2gL(1 + \cos \theta) \implies v^2 = gL(1 - 2 \cos \theta)$$

Condition for string slacking ($T = 0$):

$$mg \cos \theta = \frac{mv^2}{L} \implies v^2 = gL \cos \theta$$

Equating both:

$$gL \cos \theta = gL(1 - 2 \cos \theta) \implies 3 \cos \theta = 1 \implies \cos \theta = \frac{1}{3}$$

For $u = \sqrt{3gL}$, $\cos \theta = 0.5 \implies \theta = 60^\circ$.

Sol 4. Option (D)

Total vertical fall = $h + x$.

By conservation of mechanical energy ($\Delta K = 0$):

$$mg(h + x) = \frac{1}{2}kx^2$$

$$1(10)(0.45 + x) = \frac{1}{2}(1000)x^2 \implies 500x^2 - 10x - 4.5 = 0$$

$$100x^2 - 2x - 0.9 = 0 \implies x = \frac{2 + \sqrt{4 - 4(100)(-0.9)}}{200} = \frac{2 + 19.07}{200}$$

Sol 5. Option (C)

Turning points occur where kinetic energy is zero ($K = 0 \implies U(x) = E$):

$$2x^2 - 8x + 10 = 18 \implies 2x^2 - 8x - 8 = 0 \implies x^2 - 4x - 4 = 0$$

For $E = 18$ J: $2(x - 2)^2 + 2 = 18 \implies (x - 2)^2 = 8 \implies x = 2 \pm 2\sqrt{2}$.

If $U(x) = x^2 - 4x + 10 = 18 \implies (x - 2)^2 = 16 \implies x = 2 \pm 4 \implies x \in [-2, 6]$ m.

Sol 6. Option (B)

Let x be the stretch in the cord. Total drop = $L + x$.

By conservation of energy:

$$mg(L + x) = \frac{1}{2}kx^2 \implies kx^2 - 2mgx - 2mgL = 0$$

$$x = \frac{2mg + \sqrt{4m^2g^2 + 8kmgL}}{2k} = \frac{mg}{k} + \sqrt{\frac{m^2g^2}{k^2} + \frac{2mgL}{k}}$$

Total displacement from O is $y = L + x$.

Sol 7. Option (B)

To complete a vertical loop of radius R , the minimum speed at the lowest point must be $v_{\text{bottom}} = \sqrt{5gR}$.

By conservation of energy:

$$mgh = \frac{1}{2}mv_{\text{bottom}}^2 = \frac{1}{2}m(5gR) \implies h = \frac{5}{2}R = 2.5R$$

Sol 8. Option (C)

Normal force consists of two parts: 1. Weight of the fallen chain: $W = \left(\frac{M}{L}y\right)g$. 2. Reaction due to momentum change of arriving links at speed $v = \sqrt{2gy}$:

$$F_{\text{impact}} = \frac{dm}{dt}v = \left(\frac{M}{L}v\right)v = \frac{Mv^2}{L} = \frac{M(2gy)}{L} = \frac{2Mgy}{L}$$

Total force:

$$N = W + F_{\text{impact}} = \frac{Mgy}{L} + \frac{2Mgy}{L} = \frac{3Mgy}{L}$$

Sol 9. Option (A)

Maximum height: $H = \frac{u^2 \sin^2 \theta}{2g}$.

At height $h = H/2 = \frac{u^2 \sin^2 \theta}{4g}$:

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgh \Rightarrow v^2 = u^2 - 2g \left(\frac{u^2 \sin^2 \theta}{4g} \right)$$

$$v^2 = u^2 \left(1 - \frac{\sin^2 \theta}{2} \right) \Rightarrow v = u \sqrt{1 - \frac{\sin^2 \theta}{2}}$$

Sol 10. Option (A)

At maximum compression, both blocks move with the common velocity:

$$V_{\text{cm}} = \frac{m_1 v_0}{m_1 + m_2}$$

Loss in kinetic energy goes into elastic energy of the spring:

$$\frac{1}{2}\mu v_0^2 = \frac{1}{2}kx_{\text{max}}^2 \Rightarrow x_{\text{max}} = v_0 \sqrt{\frac{\mu}{k}} = v_0 \sqrt{\frac{m_1 m_2}{(m_1 + m_2)k}}$$

Sol 11. Option (B)

Speed at lowest point:

$$\frac{1}{2}mv^2 = mgL(1 - \cos 60^\circ) = mgL \left(1 - \frac{1}{2} \right) = \frac{1}{2}mgL \Rightarrow v^2 = gL$$

Tension at lowest point:

$$T = mg + \frac{mv^2}{L} = mg + \frac{m(gL)}{L} = 2mg$$

Sol 12. Option (A)

Potential energy: $U(x) = \frac{a}{x^2} - \frac{b}{x}$.

Equilibrium position: $\frac{dU}{dx} = -\frac{2a}{x^3} + \frac{b}{x^2} = 0 \Rightarrow x_0 = \frac{2a}{b}$.

Depth of well:

$$U(x_0) = \frac{a}{(2a/b)^2} - \frac{b}{(2a/b)} = -\frac{b^2}{4a}$$

Escape speed v_{esc} requires total energy $E \geq 0$:

$$\frac{1}{2}mv_{\text{esc}}^2 + U(x_0) = 0 \Rightarrow \frac{1}{2}mv_{\text{esc}}^2 = \frac{b^2}{4a} \Rightarrow v_{\text{esc}} = \frac{b}{\sqrt{2ma}}$$

Sol 13. Option (B)

Mechanical energy ($E = K + U$) is strictly conserved if and only if no non-conservative forces (like friction, viscous drag, inelastic deformation) do net work on the system ($W_{\text{nc}} = 0$).

Sol 14. Option (A)

Smooth incline (pure translation):

$$mgh = \frac{1}{2}Mv_1^2 \Rightarrow v_1 = \sqrt{2gh}$$

Rough incline (pure rolling of solid sphere with $I = \frac{2}{5}MR^2$):

$$mgh = \frac{1}{2}Mv_2^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv_2^2 \left(1 + \frac{2}{5} \right) = \frac{7}{10}Mv_2^2 \Rightarrow v_2 = \sqrt{\frac{10}{7}gh}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{10/7}{2}} = \sqrt{\frac{5}{7}}$$

Sol 15. Option (A)

Both blocks move together with common acceleration until natural length is reached:

$$\frac{1}{2}kx_0^2 = \frac{1}{2}(m+2m)v^2 \Rightarrow \frac{1}{2}kx_0^2 = \frac{3}{2}mv^2 \Rightarrow v = x_0 \sqrt{\frac{k}{3m}}$$

Sol 16. Option (C)

Height fallen from top to bottom = $2R$.

By conservation of energy:

$$\frac{1}{2}mv^2 = mg(2R) \Rightarrow v^2 = 4gR$$

Normal reaction at bottom:

$$N - mg = \frac{mv^2}{R} \Rightarrow N = mg + \frac{m(4gR)}{R} = 5mg$$

Sol 17. Option (A)

Tension is purely radial, so angular momentum is conserved:

$$mv_1 r_1 = mv_2 r_2 \Rightarrow v_2 = v_1 \left(\frac{r_1}{r_2} \right)$$

By Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}mv_1^2 \left(\frac{r_1^2}{r_2^2} - 1 \right)$$

Sol 18. Option (A)

By energy conservation: $\frac{1}{2}mv^2 = mgh \Rightarrow v^2 = 2gh$.

Radius of curvature at $x = 0$ for $y = kx^2$:

$$R_c = \frac{1}{y''(0)} = \frac{1}{2k}$$

Centripetal equation at bottom:

$$N - mg = \frac{mv^2}{R_c} = \frac{m(2gh)}{1/(2k)} = 4kmgh \Rightarrow N = mg(1 + 4kh)$$

Sol 19. Option (A)

Parallel stiffness: $k_{\text{eff}} = 2k$.

Total energy: $E = \frac{1}{2}k_{\text{eff}}x_0^2 = kx_0^2$.

At $x = x_0/2$:

$$U = \frac{1}{2}(2k) \left(\frac{x_0}{2} \right)^2 = \frac{1}{4}kx_0^2$$

$$K = E - U = kx_0^2 - \frac{1}{4}kx_0^2 = \frac{3}{4}kx_0^2$$

Sol 20. Option (B)

The new radius of curvature around the nail is $r = L - d$.

To loop around the nail, the speed at the top of the circle must be at least $\sqrt{gr}$.

Height of top of new loop above lowest point = $2r = 2(L - d)$.

Energy conservation gives:

$$L \geq \frac{5}{2}(L - d) \Rightarrow \frac{5}{2}d \geq \frac{3}{2}L \Rightarrow d \geq \frac{3}{5}L$$